NON-MF GROUPS

SAUERS

Some groups are not MF. For example, $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ is a simple property-(T) group, and every homomorphism from H to an MF group is trivial. We also construct a two-generated, finitely presented, torsion-free property-(T) group with the same rigidity, and a sofic non-MF affine–Clifford group W that is an extension of $\mathbb{Z}$ by a locally residually finite group. The common mechanism is a normal property-(T) subgroup contained in the normal closure of commutators arising from a one-sided compression. Consequently, $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF, while the canonical trace on $C^*_{\max}(W)$ is amenable but not quasidiagonal.

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1. Introduction

The counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)),$$

the elementary group of 12×12 matrices over the binary Leavitt algebra. Khanh–Thanh show that H is isomorphic to the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4]. OpenAI used this group to construct the first nonsofic group [OAI, Chapter 3].

We work with countable groups throughout. A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

In other words, G is MF if there are finite unitary matrices, one for each element of G, that multiply correctly up to an error tending to zero in operator norm while no nonidentity element converges to the identity matrix. For countable G, the separation may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a group that is not MF as defined here is not MF in that convention either. Equivalently, G embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

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Here and below, products of C^*-algebras mean bounded products, and $\bigoplus$ denotes the ideal of norm-null sequences. The quotient $\mathcal{Q}_{\mathbf{d}}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$, since $\prod_n M_{d_n}(\mathbb{C})$ is its multiplier algebra. We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. A separable C^*-algebra is MF if it embeds in a norm matrix corona [BK].

We use the group commutator convention $[g,h] = ghg^{-1}h^{-1}$. The groups here fail to be MF because they contain a property-(T) subgroup that is conjugated into a proper subgroup of itself. For $L \leq G$ put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}.$$

An element c of the centralizer $C_G(L)$ commutes with L, so ucu^{-1} commutes with uLu^{-1} but need not commute with the rest of L. The normal subgroup of G generated by the commutators of all such elements ucu^{-1} with elements of L is

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then every homomorphism from G to an MF group is trivial on K. In particular, a nontrivial such K makes G non-MF, and if G has property* (T) *and $\mathfrak{D}_G(L) = G$, then every homomorphism from G to an MF group is trivial.*

Asymptotic multiplicativity in operator norm makes the conjugation maps $\mathrm{Ad}(V_n(g))\colon x \mapsto V_n(g)xV_n(g)^*$ an asymptotically multiplicative family of unitaries on $M_{d_n}(\mathbb{C})$ with its normalized Hilbert–Schmidt inner product. Here $\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \leq 2\|A - B\|$, with the norm on the left taken in the operator algebra of this Hilbert space. Passing to a Hilbert ultraproduct turns these maps into a genuine unitary representation of G, to which property (T) applies. Let U be the class of $(\mathrm{Ad}(V_n(u)))$ and P the Kazhdan projection of L for the conjugation maps, both in the norm matrix corona with coordinate sizes d_n^2. Then $U^*PU \leq P$, because $uLu^{-1} \leq L$ has fewer constraints and more fixed vectors, and stable finiteness of that corona forces $U^*PU = P$. So conjugation by u preserves the Hilbert–Schmidt asymptotic commutant of L, and every element of $\mathfrak{D}_G(L)$ tends to 1 in Hilbert–Schmidt norm in every operator norm asymptotic representation (Corollary 2.5). Property (T) of K turns this into triviality in operator norm. A corona homomorphism nontrivial on K compresses to a corner where the Kazhdan projection of K vanishes. On that corner, Hilbert–Schmidt triviality makes every ultralimit of normalized traces equal to the trivial character of K, which takes the value 1 at the Kazhdan projection (Theorem 2.7).[1]

The stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is a compressor. The group H realizes the compression inside a simple group, where one nontrivial commutator $[ucu^{-1}, \ell]$ normally generates everything. Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra, the unital $\mathbb{F}_2$-algebra

[1] A Hilbert–Schmidt bound on the multiplicative defect does not give an operator norm bound on the conjugation maps, so this argument does not apply to sofic or hyperlinear approximations.

generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property* (T)*, and every homomorphism from H to an MF group is trivial. So H is not MF, and $C^*_{\mathrm{r}}(H)$ is separable and stably finite but not MF, while $C^*_{\max}(H)$ is not even finite: it contains a proper isometry.*

Theorem C (a torsion-free finitely presented example)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property* (T) *such that every homomorphism from Q to an MF group is trivial. In particular, no nontrivial quotient of Q is MF.*

Section 5 builds Q from Fournier-Facio's torsion-free property-(T) group [FF, §2] and Hull's small cancellation theorem [Hu].

Theorem D (an amenable nonquasidiagonal trace)**.** *There is a sofic group $W = K \rtimes \mathbb{Z}$, with K locally residually finite, that is not MF. The canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal.*

Section 4 constructs W from the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and Clifford lamps. The group W is sofic because locally residually finite groups are sofic and soficity passes to extensions with amenable quotient [ES, Theorem 1]. The trace answers the question of Brown [Br, discussion preceding Proposition 3.5.1] and of Schafhauser, Tikuisis and White [STW, Problem X(1)] whether every amenable trace is quasidiagonal. Since K is a direct limit of residually finite groups, it is MF [Kor, Corollary 10 and Proposition 13]. So MF groups are not closed under semidirect products with $\mathbb{Z}$, which answers a question of Korchagin [Kor, remark following Proposition 12].

Relation to prior work. Blackadar and Kirchberg introduced MF algebras and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. Whether every separable stably finite C^*-algebra is MF, the MF problem [GH, §6], was answered negatively through the failure of the Connes embedding problem [JNVWY][GH, Proposition 6.1 and Remark 6.2]. Theorem B gives a counterexample among reduced group C^*-algebras. Whether every countable group is MF was a longstanding open problem [BDL].

Theorem A uses the configuration of the nonsofic construction [OAI, Proposition 2.3]: a property-(T) subgroup L conjugated into itself by an element u, and an element c commuting with L whose conjugate ucu^{-1} does not. There, soficity and the rigidity theorems of Kun [Ku16] and Kun–Thom [KT19] force the centralizing subgroup to be locally embeddable into finite groups, and a copy of Thompson's group V gives the contradiction. Here the rigidity is the Kazhdan projection of L in a norm matrix corona, which stable finiteness of the corona forces to commute with the compressor, and the contradiction is a commutator $[ucu^{-1}, \ell] \neq 1$ that every operator norm asymptotic representation sends to 1 in Hilbert–Schmidt norm. The group H is isomorphic to the unit group of the binary Leavitt algebra, the group of [OAI], and Theorem C reuses Fournier-Facio's configuration [FF]. Bachner–Dogon–Lubotzky showed that for groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. The compression relation replaces stability.

For a property-(T) group Γ with a proper self-embedding and its ascending HNN extension G, Eckhardt [Ec] shows that $C^*_{\max}(G)$ is not finite, as in Proposition 2.9, and that the wreath product $(\bigoplus_{G/\Gamma} \mathbb{Z}/2\mathbb{Z}) \rtimes G$ is not MF. In Section 4 the lamps satisfy Clifford relations, and by these relations a central involution lies in the normal subgroup (1), so Theorem A applies directly.

Leavitt introduced the algebras $L_K(1,n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then gives simplicity of H [Pre], and property (T) is Ershov–Jaikin-Zapirain's theorem [EJZ, Theorem 1.1].

2. One-sided compression

2.1. **Corona homomorphisms.** If G is countable, the image of a corona homomorphism from G is a countable subgroup of a corona unitary group, so MF. Every MF group embeds in the unitary group of a norm matrix corona.

Lemma 2.1. *Let G be countable and $K \leq G$. Every homomorphism from G to an MF group is trivial on K if and only if every corona homomorphism from G is trivial on K.*

Proof. The image of a corona homomorphism is a countable MF group. Conversely, compose a homomorphism to an MF group with a corona embedding of its image; the resulting corona homomorphism has the same kernel. □

2.2. **Kazhdan transport in normalized Hilbert–Schmidt norm.** For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a group G is a sequence of maps $V_n \colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

The elements g with $\|V_n(g) - 1\|_2 \to 0$ form a normal subgroup of G, because $\|a\|_2 \leq \|a\|$, the Hilbert–Schmidt norm is invariant under adjoints and unitary conjugation, and $\|V_n(g^{-1}) - V_n(g)^*\| \to 0$:

$$\|V_n(gh) - 1\|_2 \leq \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1),$$

$$\left\|V_n(g^{-1}) - 1\right\|_2 = \|V_n(g) - 1\|_2 + o(1), \qquad \left\|V_n(ghg^{-1}) - 1\right\|_2 = \|V_n(h) - 1\|_2 + o(1).$$

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|V_n(\ell)x_n - x_nV_n(\ell)\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C})/\bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. If $v^*v = 1$ in A and (x_n) is a bounded lift of v, then $\|x_n^*x_n - 1\| \to 0$, so for all large n the matrix $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary and $\|u_n - x_n\| \to 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v, and $vv^* = 1$. So A is finite, and so is $M_k(A)$, which is the corona with matrix sizes km_n. Finally, if $p \le q$ are equivalent projections in a stably finite algebra B and w is a partial isometry with $w^*w = q$ and $ww^* = p$, then $w \in qBq$ is an isometry in the finite corner qBq, so $p = ww^* = q$. □

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. Then P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$ because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^*\operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, and the range projection U^*PU is dominated by P in $B(\mathcal{H})$, and then in B by faithfulness. □

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\mathrm{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and a free ultrafilter ω, and let $\mathcal{K}_\omega$ be the ultraproduct of the Hilbert spaces $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$. For unitaries A, B,

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|\,,$$

where the norm on the left is the operator norm on this Hilbert space, so the maps $\mathrm{Ad}(V_n(g))$ are asymptotically multiplicative in operator norm, and

$$\sigma(g) = \lim_\omega \mathrm{Ad}(V_n(g))$$

is a unitary representation of G on $\mathcal{K}_\omega$ with $\xi = [x_n]_\omega \in \operatorname{Fix}\sigma(L)$. The same estimate makes

$$\widetilde{\sigma}(g) = [\operatorname{Ad}(V_n(g))]_n \in \mathcal{U}(\mathcal{B}), \qquad \mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})),$$

a homomorphism, where $B(M_{d_n}(\mathbb{C}))$ denotes the operators on the Hilbert–Schmidt Hilbert space. Coordinatewise action induces a representation $\pi_\omega \colon \mathcal{B} \to B(\mathcal{K}_\omega)$, which we suppress from the notation. Moreover, $\mathcal{B}$ is a norm matrix corona with coordinate sizes d_n^2 after a choice of matrix units. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L [AW] under $C^*_{\max}(L) \to \mathcal{B}$. On $\mathcal{K}_\omega$ its range is $\operatorname{Fix}\sigma(L)$. Put $U = \widetilde{\sigma}(u)$. Since $uLu^{-1} \leq L$,

$$U\widetilde{\sigma}(L)U^* \subseteq \widetilde{\sigma}(L), \qquad \text{so} \qquad U^*PU \leq P$$

by Lemma 2.3, and

$$U^*PU = P$$

by Lemma 2.2, since the two projections are unitarily equivalent. So U commutes with P, and

$$U\xi,\ U^*\xi \in \operatorname{Fix}\sigma(L), \qquad \text{that is,}$$

$$\lim_\omega \left\| V_n(\ell)V_n(u)^{\pm 1}x_nV_n(u)^{\mp 1} - V_n(u)^{\pm 1}x_nV_n(u)^{\mp 1}V_n(\ell) \right\|_2 = 0 \qquad (\ell \in L).$$

Since ω was arbitrary, the limits hold along the full sequence: a subsequence staying above some $\varepsilon > 0$ would lie in a free ultrafilter and give a nonzero ultralimit. So $\operatorname{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$. □

Corollary 2.5. *Let $L \leq G$ have property* (T) *and let (V_n) be an operator norm asymptotic representation of G. Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \qquad (d \in \mathfrak{D}_G(L)).$$

Proof. Let $u \in \operatorname{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, and write $\mathcal{C}_2 = \mathcal{C}_2(V, L)$. Then

$$(V_n(c)) \in \mathcal{C}_2 \implies (V_n(u)V_n(c)V_n(u)^*) \in \mathcal{C}_2 \implies (V_n(ucu^{-1})) \in \mathcal{C}_2,$$

the first membership because c commutes with L, the first implication by Theorem 2.4, and the second because $\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \to 0$. So $\|V_n(\ell)V_n(ucu^{-1}) - V_n(ucu^{-1})V_n(\ell)\|_2 \to 0$. Hence, by asymptotic multiplicativity, $\|V_n([ucu^{-1}, \ell]) - 1\|_2 \to 0$. The elements with this property form a normal subgroup of G, so it contains $\mathfrak{D}_G(L)$. □

2.3. **From Hilbert–Schmidt to operator norm.** A corona homomorphism ρ cannot simply be restricted to a corner: a projection q commuting with $\rho(G)$ has lifts q_n that only asymptotically commute with chosen unitary lifts $U_n(g)$ of $\rho(g)$, so the compressions $q_nU_n(g)q_n$ are only approximately unitary in the matrix corners.

Lemma 2.6 (central corona corners)**.** *Let $\rho \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections*

$q_n \in M_{d_n}(\mathbb{C})$, *integers* $r_n = \operatorname{rank}(q_n)$, *unitary identifications* $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, *and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$ *such that, in the corona over the retained coordinates, the class of* $(J_nW_n(g)J_n^*)$ *is the coordinate restriction of* $q\rho(g)$*, a unitary of the corner* $q\mathcal{Q}_{\mathbf{d}}q$ *with unit* q*.*

Proof. Lift q to projections q_n by spectral rounding, and lift each $\rho(g)$ to unitaries $U_n(g)$ by polar correction, with $U_n(1) = I_{d_n}$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain those coordinates and put $r_n = \operatorname{rank}(q_n)$. The commutation relation in the corona gives $\|q_nU_n(g) - U_n(g)q_n\| \to 0$ for each g, so [BDL, Proposition 2.4] with $p = \infty$ gives an operator norm asymptotic representation $\widetilde{W}_n$ of G by unitaries of the corner $q_nM_{d_n}(\mathbb{C})q_n$ with $\left\|\widetilde{W}_n(g) - q_nU_n(g)q_n\right\| \to 0$. Replacing $\widetilde{W}_n(1)$ by q_n changes it by $o(1)$. Choose unitaries $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$ and put $W_n(g) = J_n^*\widetilde{W}_n(g)J_n$. Conjugation by J_n preserves operator norms and multiplicative defects, so (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$, and $\|J_nW_n(g)J_n^* - q_nU_n(g)q_n\| \to 0$ identifies the class of $(J_nW_n(g)J_n^*)$ with the coordinate restriction of $q\rho(g)$. $\square$

Theorem 2.7 (normal Kazhdan subgroups)**.** *Let* G *be countable and let* $K \trianglelefteq G$ *have property* (T)*. If every operator norm asymptotic representation* (V_n) *of* G *satisfies* $\|V_n(k) - 1\|_2 \to 0$ *for all* $k \in K$*, then every corona homomorphism from* G *is trivial on* K*.*

Proof. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Let $e_K \in C^*_{\max}(K)$ be the Kazhdan projection of K, and let $p \in \mathcal{Q}_{\mathbf{d}}$ be its image under the homomorphism $C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{d}}$ induced by $\Theta|_K$. In any faithful representation of the corona, p is the projection onto the K-fixed vectors. Since $gKg^{-1} = K$,

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G),$$

and $q = 1 - p$ is nonzero, because $p = 1$ would make every element of K act trivially. Lemma 2.6 gives an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$. Let $\pi\colon C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{r}}$ be the homomorphism induced by $\widehat{\Theta}|_K$. Since q commutes with $\Theta(K)$, the map $a \mapsto q\Theta(a)$ is a homomorphism on $C^*_{\max}(K)$ agreeing with π on K, so $\pi(e_K)$ is the coordinate restriction of $qp = 0$.

Fix a free ultrafilter ω. Since $|\operatorname{tr}_{r_n}(a_n)| \leq \|a_n\|$, the formula

$$T_\omega([a_n]) = \lim_\omega \operatorname{tr}_{r_n}(a_n)$$

defines a tracial state on $\mathcal{Q}_{\mathbf{r}}$. The hypothesis applies to (W_n) itself, so $\|W_n(k) - I_{r_n}\|_2 \to 0$ for $k \in K$ in the normalized Hilbert–Schmidt norm of $M_{r_n}(\mathbb{C})$. Since $|\operatorname{tr}_{r_n}(W_n(k)) - 1| \leq \|W_n(k) - I_{r_n}\|_2$, it follows that $T_\omega(\pi(k)) = 1$ for every $k \in K$. Then the states $T_\omega \circ \pi$ and the trivial character of K agree on the canonical unitaries, whose span is dense in $C^*_{\max}(K)$, and therefore on all of $C^*_{\max}(K)$. The trivial character takes the value 1 at e_K, while $\pi(e_K) = 0$. So $0 = T_\omega(\pi(e_K)) = 1$, a contradiction. $\square$

Proof of Theorem A. By Corollary 2.5, every operator norm asymptotic representation of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for $k \in K \leq \mathfrak{D}_G(L)$. Theorem 2.7 then makes every corona homomorphism trivial on K, and Lemma 2.1 makes every homomorphism to an MF group trivial on K. If $K \neq 1$, then G does not embed in an MF group, so G is not MF. The case $K = G = \mathfrak{D}_G(L)$ gives the last assertion. □

2.4. **The maximal group C^*-algebra.** Lemma 2.3 also applies inside $C^*_{\max}(G)$ itself, where a strict compression produces a proper isometry.

Lemma 2.8 (strictly dominated projections)**.** *Let A be a unital C^*-algebra containing a projection p and a unitary u with $p < upu^*$. Then $s = pu^* + (1 - upu^*)$ is an isometry that is not a unitary, so A is not stably finite and has no faithful tracial state.*

Proof. Put $q = upu^*$, so that $pq = qp = p$, $up(1-q) = 0$, and $pu^*(1-q) = pu^* - pu^*upu^* = 0$. Then

$$\begin{aligned} s^*s &= upu^* + up(1-q) + (1-q)pu^* + (1-q) = q + (1-q) = 1, \\ ss^* &= pu^*up + pu^*(1-q) + (1-q)up + (1-q) \\ &= p + (1-q) = 1 - (q-p) \neq 1. \end{aligned}$$

A unital C^*-algebra containing a nonunitary isometry is not finite, and neither is any matrix algebra over it, since $\operatorname{diag}(s, 1, \ldots, 1)$ is a nonunitary isometry in $M_k(A)$. A tracial state τ has $\tau(q-p) = \tau(s^*s) - \tau(ss^*) = 0$ with $q - p$ a nonzero positive element, so no tracial state on A is faithful. □

Proposition 2.9 (proper one-sided compression)**.** *Let G be a countable group containing a property-*(T) *subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ contains a proper isometry. In particular, $C^*_{\max}(G)$ is not stably finite, admits no faithful tracial state, and is neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the image of the Kazhdan projection of Γ [AW] under the canonical map $C^*_{\max}(\Gamma) \to C^*_{\max}(G)$, and let u be the canonical unitary of t. Conjugation by u implements the isomorphism $\Gamma \to t\Gamma t^{-1}$ on canonical unitaries, so Lemma 2.3, applied with $U = u$, gives $P \leq uPu^*$ with uPu^* the image of the Kazhdan projection of $t\Gamma t^{-1}$. The inequality is strict: in the quasi-regular representation of G on $\ell^2(G/t\Gamma t^{-1})$, the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections differ. Lemma 2.8 then supplies the proper isometry and rules out stable finiteness and faithful traces. MF algebras are stably finite, and a separable residually finite-dimensional C^*-algebra is MF [BK]. □

3. The binary Leavitt group

The Leavitt algebra $R = L_{\mathbb{F}_2}(1,2)$ is built so that one copy of the free module R is isomorphic to two copies: the relations (2) say that s_0, s_1 embed two copies of R side by side and t_0, t_1 read them back. So a matrix A over R can be copied into one of the two halves, with the identity on the other half, and this compression is an injective multiplicative map that fixes the identity. Its restriction to the unit group is an injective group homomorphism. Inside a large enough elementary group the compression is realized by conjugation, which gives a compressor τ for a copy L of $\mathrm{EL}_3(R)$; one commutator computation gives a nontrivial element of $\mathfrak{D}_H(L)$, and simplicity of H turns it into all of H.

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1,$$

so that $p + q = 1$ and $t_1 q s_1 = 1$, and in particular $q \neq 0$. We use indices $0, \dots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ of matrices $\mathrm{diag}(A, I_9)$. Define $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0 A t_0, \tag{3}$$

where $s_0 A t_0$ is the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0 q = 0$, and $t_0 s_0 = 1$ show that Ψ is multiplicative, $\Psi(I_3) = (p+q)I_3 = I_3$, and $t_0 \Psi(A) s_0 = A$. Thus it restricts to a group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$, and the recovery identity makes that homomorphism injective. Put

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix};$$

block multiplication with (2) gives $Y = X^{-1}$ and $X\,\mathrm{diag}(A, I_3) Y = \mathrm{diag}(\Psi(A), I_3)$. In twelve coordinates the compressor is elementary by Whitehead's lemma: for

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R) \tag{4}$$

one has

$$\tau\ \mathrm{diag}(A, I_9)\, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{5}$$

Lemma 3.1. *The matrix τ defined in (4) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. By the Whitehead lemma [Mil, §3], $\mathrm{diag}(X, X^{-1})$ is a product of block-triangular matrices $\left(\begin{smallmatrix} I & B \\ 0 & I \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I & 0 \\ B & I \end{smallmatrix}\right)$ with $B \in M_6(R)$, and each of these is the product of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, since distinct off-diagonal matrix units have product zero. □

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \le H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \le L. \tag{6}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring, so the theorem of Ershov and Jaikin-Zapirain gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$. For $i \neq j$ and $a \in R$, equations (5) and (3) give $\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L$, and these elementary matrices generate L. □

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], so it is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $N \le \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$, Preusser's sandwich theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ with $\mathrm{EL}_{12}(R,I) \le N \le C_{12}(R,I)$, where $\mathrm{EL}_{12}(R,I)$ is the relative elementary subgroup and $C_{12}(R,I)$ is the full congruence subgroup. Simplicity of R gives $I=0$ or $I=R$, and $I=R$ gives $N=H$. If $I=0$, then N centralizes every $e_{ij}(a)$: centralizing $e_{ij}(1)$ for all $i \neq j$ forces $g=\lambda I_{12}$ with $\lambda \in R^\times$, and centralizing $e_{ij}(a)$ forces $\lambda \in Z(R)^\times$. Since $Z(R)=\mathbb{F}_2$ [AC, Corollary 4.3], $N=1$. Finally, $e_{01}(1) \neq 1$. $\square$

Proposition 3.4. *Let*

$$c=e_{34}(1), \qquad \ell=e_{12}(1), \qquad d=[\tau c\tau^{-1},\ell]$$

in H. Then

$$c \in C_H(L), \qquad d=e_{02}(q) \neq 1, \qquad \langle\!\langle d\rangle\!\rangle_H=H.$$

Proof. The Steinberg relations give $[e_{ij}(a),e_{34}(1)]=1$ for $i,j\in\{0,1,2\}$, and these matrices generate L, so $c \in C_H(L)$. Since $\tau=\mathrm{diag}(X,Y)$ and E_{34} is supported in the first six coordinates, $\tau E_{34}\tau^{-1}=\mathrm{diag}(XE_{34}Y,0)$ with E_{34} read in $M_6(R)$, and with $\varepsilon_0,\varepsilon_1,\varepsilon_2$ the standard basis of R^3,

$$XE_{34}Y=\begin{pmatrix} s_1t_0\varepsilon_0 \\ t_1\varepsilon_0\end{pmatrix}\begin{pmatrix} s_0t_1\varepsilon_1^{\mathsf T} & s_1\varepsilon_1^{\mathsf T}\end{pmatrix}=\begin{pmatrix} s_1t_1E_{01} & 0 \\ 0 & E_{01}\end{pmatrix}$$

by (2). So

$$\tau c\tau^{-1}=1+qE_{01}+E_{34}=e_{01}(q)e_{34}(1).$$

The second factor commutes with ℓ, and $[e_{ij}(a),e_{jk}(b)]=e_{ik}(ab)$ gives $d=e_{02}(q)$, which is nontrivial because $q\neq 0$. Since H is simple, d normally generates H. $\square$

Proof of Theorem B. Proposition 3.2 gives property (T), which implies finite generation [BHV, Theorem 1.3.1]. The containment (6), the centrality of c relative to L, and Proposition 3.4 put d in $\mathfrak{D}_H(L)$, and $\langle\!\langle d\rangle\!\rangle_H=H$ gives $\mathfrak{D}_H(L)=H$. So Theorem A, applied with $K=H$, makes every homomorphism from H to an MF group trivial, and Proposition 3.3 makes H nontrivial and simple. In particular H is not MF. The algebra $C^*_{\mathrm r}(H)$ is separable, and stably finite because its canonical trace is faithful. If an embedding $e\colon C^*_{\mathrm r}(H)\to\mathcal{Q}_{\mathbf d}$ existed and $p=e(1)$, then $h\mapsto e(\lambda_h)+(1-p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf d})$, contradicting that H is not MF. Finally, the compression (6) is strict: by (5) and (3), every off-diagonal entry of a matrix in $\tau L\tau^{-1}$ has the form s_0at_0, and

$$t_1(s_0at_0)s_1=0\neq 1=t_1\cdot 1\cdot s_1,$$

so $e_{12}(1)\in L\setminus \tau L\tau^{-1}$. Proposition 2.9, applied to $L\le H$ and τ, puts a proper isometry in $C^*_{\max}(H)$. $\square$

4. An amenable nonquasidiagonal trace

Brown observed that it was unknown whether every amenable trace on a C^*-algebra is quasidiagonal [Br, discussion preceding Proposition 3.5.1]. We use the following sequential definitions for a separable unital C^*-algebra A; all applications below are in this scope. A tracial state τ is *amenable* if there are u.c.p. maps $\phi_n\colon A \to M_{d_n}(\mathbb{C})$ such that

$$\|\phi_n(ab) - \phi_n(a)\phi_n(b)\|_2 \longrightarrow 0, \qquad \operatorname{tr}_{d_n}(\phi_n(a)) \longrightarrow \tau(a) \quad (a, b \in A).$$

It is *quasidiagonal* if the first limit holds in operator norm instead of normalized Hilbert–Schmidt norm, with the same trace convergence.

For a countable group G, let u_g denote the canonical unitary in $C^*_{\max}(G)$. The canonical trace τ_G is determined by

$$\tau_G(u_g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Theorem 4.1 (canonical-trace obstruction)**.** *Let G be a countable group. If G is not MF, then its canonical trace τ_G on $C^*_{\max}(G)$ is not quasidiagonal. Consequently, if τ_G is amenable, then it is amenable and not quasidiagonal.*

Proof. Suppose that τ_G is quasidiagonal, and let ϕ_n be u.c.p. models as above. Operator-norm multiplicativity makes both $\phi_n(u_g)^*\phi_n(u_g)$ and $\phi_n(u_g)\phi_n(u_g)^*$ converge to 1. Polar correction therefore gives unitaries $V_n(g)$, with $V_n(1) = 1$, such that

$$\|V_n(g) - \phi_n(u_g)\| \longrightarrow 0.$$

The maps $V_n\colon G \to \mathcal{U}(d_n)$ are asymptotically multiplicative in operator norm. If $g \neq 1$, then $\operatorname{tr}_{d_n}(\phi_n(u_g)) \to \tau_G(u_g) = 0$. Hence $V_n(g)$ cannot converge to 1 in operator norm, since that would force $\operatorname{tr}_{d_n}(\phi_n(u_g)) \to 1$. Thus (V_n) is an MF approximation of G, contrary to the hypothesis. The final assertion merely combines this obstruction with amenability. □

A group N is *locally residually finite* if each of its finitely generated subgroups is residually finite.

Proposition 4.2 (extensions of $\mathbb{Z}$ by locally residually finite groups)**.** *Suppose that G is countable and*

$$G \cong N \rtimes_\beta \mathbb{Z},$$

where N is locally residually finite. Then τ_G is amenable.

Proof. Write $g = x_g t^{k_g}$, where $x_g \in N$, $k_g \in \mathbb{Z}$, and $t = (1, 1)$, so $txt^{-1} = \beta(x)$ for $x \in N$. Call k_g the height of g. Fix a finite set $F \subseteq G$ and a positive integer L, and put $I_L = \{0, \dots, L-1\}$. Define

$$a_{g,j} = t^{-(j+k_g)} g t^j = \beta^{-(j+k_g)}(x_g) \in N, \qquad g \in F, \quad j \in I_L,$$

and put $N_{F,L} = \langle a_{g,j} : g \in F,\ j \in I_L \rangle$. This finitely generated subgroup of N is residually finite. Taking a finite product of separating quotients, and restricting to its image, gives a finite quotient

$$\theta_{F,L}\colon N_{F,L} \longrightarrow Q_{F,L}$$

that separates every nonidentity $a_{g,j}$. Let $H_{F,L} \leq G$ be the image of $\ker \theta_{F,L} \leq N_{F,L}$ under the inclusion $N \hookrightarrow G$. Choose a representative $r_q \in N_{F,L}$ for each $q \in Q_{F,L}$ and set

$$S_{F,L} = \{t^j r_q H_{F,L} : j \in I_L,\ q \in Q_{F,L}\} \subseteq G/H_{F,L}.$$

These cosets are pairwise distinct. Let $\lambda_{F,L}$ be the quasi-regular representation of G on $\ell^2(G/H_{F,L})$ and let $P_{F,L}$ be the projection onto $\ell^2(S_{F,L})$. Put

$$\phi_{F,L}(b) = P_{F,L}\lambda_{F,L}(b)P_{F,L}\big|_{\ell^2(S_{F,L})} \qquad (b \in C^*_{\max}(G)).$$

This map is u.c.p., being the compression of a unitary representation. For $g \in F$ and $j + k_g \in I_L$, the action satisfies

$$g(t^j r_q H_{F,L}) = t^{j+k_g} a_{g,j} r_q H_{F,L} = t^{j+k_g} r_{\theta_{F,L}(a_{g,j})q} H_{F,L}.$$

Thus the image stays in $S_{F,L}$ exactly when its height stays in I_L.

Let u_g denote the canonical unitary of g. For $g, h \in F$, the two operators $\phi_{F,L}(u_{gh})$ and $\phi_{F,L}(u_g)\phi_{F,L}(u_h)$ agree on the basis vector indexed by $s \in S_{F,L}$ whenever $h \cdot s$ remains in the window. The escaping vectors lie over the boundary levels

$$\partial_{k_h} I_L = \{j \in I_L : j + k_h \notin I_L\}.$$

On an escaping vector the compressed product is zero, while the directly compressed permutation has norm at most one. Each level has $|Q_{F,L}|$ basis vectors, so the normalized squared Hilbert–Schmidt norm satisfies

$$\|\phi_{F,L}(u_{gh}) - \phi_{F,L}(u_g)\phi_{F,L}(u_h)\|_2^2 \leq \frac{|\partial_{k_h} I_L|}{|I_L|} \leq \frac{|k_h|}{L}.$$

For fixed h the right-hand side tends to zero as $L \to \infty$.

The normalized trace of $\phi_{F,L}(u_g)$ is the fraction of points of the finite window fixed by g. If g fixed $wH_{F,L}$, then $w^{-1}gw \in H_{F,L} \leq N$, and hence $k_g = 0$. Thus a nonzero-height element fixes no selected coset. If $g \in F \setminus \{1\}$ and $k_g = 0$, fixing $t^j r_q H_{F,L}$ would give

$$r_q^{-1} a_{g,j} r_q \in H_{F,L}.$$

Since $H_{F,L} = \ker \theta_{F,L}$ is normal in $N_{F,L}$ and $r_q \in N_{F,L}$, this implies $a_{g,j} \in \ker \theta_{F,L}$. But $a_{g,j} = t^{-j} g t^j \neq 1$, contrary to separation. Thus

$$\operatorname{tr}_{L|Q_{F,L}|}(\phi_{F,L}(u_g)) = 0 \quad (g \in F \setminus \{1\}), \qquad \operatorname{tr}_{L|Q_{F,L}|}(\phi_{F,L}(1)) = 1.$$

Take a finite exhaustion $F_1 \subseteq F_2 \subseteq \cdots$ of G with $1 \in F_1$, and choose $L_n \geq n(1 + \max_{g \in F_n} |k_g|)$. The maps ϕ_{F_n,L_n} are asymptotically multiplicative in normalized Hilbert–Schmidt norm and their traces converge to τ_G on every canonical unitary. The linear span of these unitaries is dense in $C^*_{\max}(G)$; contractivity extends both limits to the whole algebra, which is precisely amenability of τ_G. □

The obstruction itself only needs a proper self-embedding of a Kazhdan group. Let Γ be a countable group with property (T), let $\alpha\colon \Gamma \to \Gamma$ be injective but not surjective, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Put

$$T_\alpha = \varinjlim(\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots), \qquad V = T_\alpha \rtimes \langle t \rangle,$$

where $tgt^{-1} = \alpha(g)$ on the level-zero copy of Γ, so that V is the ascending HNN extension of Γ along α and $T_\alpha = \bigcup_{n\geq 0} t^{-n}\Gamma t^n$. Put $X = V/\Gamma$, the left-coset space. The Clifford lamp group $\mathrm{Cl}(X)$ is presented by generators ε and c_x for $x \in X$, with relations

$$\varepsilon^2 = c_x^2 = 1, \qquad [\varepsilon, c_x] = 1, \qquad [c_x, c_y] = \varepsilon \quad (x \neq y).$$

To see that ε survives these relations, fix a total order on X and put $E = \mathbb{F}_2^{(X)}$, the finitely supported functions on X. Define a multiplication on $\mathbb{F}_2 \times E$ by

$$(a,f)(b,g) = (a+b+B(f,g),\ f+g)\,, \qquad B(f,g) = \sum_{x>y} f(x)g(y).$$

The sum is finite, and bilinearity gives $B(f,g)+B(f+g,h) = B(g,h)+B(f,g+h)$, proving associativity. The identity is $(0,0)$ and the inverse of (a,f) is $(a+B(f,f),f)$. The elements $(1,0)$ and $(0,\delta_x)$ satisfy the displayed relations: they are involutions, $(1,0)$ is central, and the commutator of two distinct $(0,\delta_x)$ is $(1,0)$. Hence the assignments $\varepsilon \mapsto (1,0)$ and $c_x \mapsto (0,\delta_x)$ define a homomorphism from the presented group, proving $\varepsilon \neq 1$. Reordering words gives forms

$$\varepsilon^a c_{x_1}\cdots c_{x_r}, \qquad a \in \mathbb{F}_2, \quad x_1 < \cdots < x_r,$$

whose images are $(a, \delta_{x_1}+\cdots+\delta_{x_r})$; thus these forms are unique. In particular, the subgroup generated by ε and the lamps on a finite set Y has order $2^{|Y|+1}$, so $\mathrm{Cl}(X)$ is locally finite and countable. Every permutation of X preserves the symmetric presentation and therefore induces an automorphism fixing ε. In particular, the permutation action of V on X acts on these lamps, and the affine–Clifford witness is

$$W = \mathrm{Cl}(X) \rtimes V. \tag{7}$$

Proposition 4.3 (Clifford obstruction from a proper self-embedding)**.** *The group W is not MF. More precisely, every homomorphism from W to an MF group kills ε.*

Proof. Let c be the lamp at the root coset. It centralizes the level-zero copy of Γ. The two cosets supporting tct^{-1} and $a(tct^{-1})a^{-1}$ are distinct precisely because $a \notin \alpha(\Gamma)$, and therefore

$$[tct^{-1}, a(tct^{-1})a^{-1}] = \varepsilon \neq 1. \tag{8}$$

Here and below the copy of Γ in W is understood. Set

$$x = tct^{-1}, \qquad y = axa^{-1}, \qquad d = [x,a].$$

Both x and y are involutions, and hence

$$d = xaxa^{-1} = xy, \qquad d^2 = (xy)^2 = [x,y] = \varepsilon.$$

Since $t \in \mathrm{Comp}_W(\Gamma)$, $c \in C_W(\Gamma)$ and $a \in \Gamma$, the element $d = [tct^{-1},a]$ is one of the generators of $\mathfrak{D}_W(\Gamma)$. Thus

$$\langle\varepsilon\rangle \leq \mathfrak{D}_W(\Gamma).$$

The subgroup $\langle\varepsilon\rangle$ is finite, central, normal and nontrivial, so it has property (T). The one-sided compression criterion applies because W is countable: the groups T_α and V, the coset space X, and $\mathrm{Cl}(X)$, whose elements have finite support, are all countable.

It therefore makes every homomorphism from W to an MF group kill ε. In particular, W is not MF. ☐

The extra hypotheses in the next proposition concern approximation of the canonical trace, not the Clifford obstruction.

Proposition 4.4 (the locally residually finite case)**.** *Suppose in addition that Γ is residually finite and that $[\Gamma : \alpha(\Gamma)] < \infty$. Then*

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha,$$

where K is locally residually finite. Consequently W is sofic but not MF, and the canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal.

Proof. Reassociating the semidirect products yields

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha.$$

For $n \geq 0$, write $\Gamma_n = t^{-n}\Gamma t^n$ for the image of the nth copy of Γ in T_α. Successive telescope copies have finite relative index because $[\Gamma : \alpha(\Gamma)] < \infty$, so all the Γ_n are commensurable. In fact Γ is commensurated by V: every element of T_α lies in some telescope copy, and conjugation by t shifts the copies. Hence the stabilizer in Γ_n of every point $v\Gamma \in V/\Gamma$ has finite index, and every Γ_n-orbit in X is finite.

Every finite subset of K involves finitely many Clifford generators and has its T_α-coordinates in one Γ_n. The union Y of the Γ_n-orbits of the corresponding sites is finite and invariant, and the original finite subset is contained in

$$C_Y \rtimes \Gamma_n,$$

where $C_Y = \langle \varepsilon, c_y : y \in Y\rangle$ is finite by the normal form above. This semidirect product is residually finite. Indeed, an element with nontrivial Γ_n-component survives in a finite quotient of Γ_n. For a nontrivial element of C_Y, let J be the kernel of the action $\Gamma_n \to \mathrm{Aut}(C_Y)$. The quotient Γ_n/J is finite, and the finite semidirect product $C_Y \rtimes (\Gamma_n/J)$ retains that element. Therefore every finitely generated subgroup of K is residually finite.

Each finitely generated subgroup of K is residually finite and hence sofic. The group K is their directed union, and $W/K \cong \mathbb{Z}$ is amenable. Soficity passes to directed unions and to extensions with amenable quotient [ES, Theorem 1], so W is sofic. Proposition 4.3 shows that it is not MF. Proposition 4.2 now makes the canonical trace amenable, and Theorem 4.1 makes it nonquasidiagonal. ☐

For a concrete instance, take

$$\bar{\Gamma} = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}) = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}^3 \right\} \leq \mathrm{GL}_4(\mathbb{Z}).$$

It is residually finite, since reduction modulo a suitable integer separates any two distinct integral matrices, and it has property (T) [BHV, Example 1.7.4(i)]. Let

$$D = \mathrm{diag}(2, 2, 2, 1), \qquad \alpha(g) = DgD^{-1},$$

so that $\alpha(v, A) = (2v, A)$. This is injective, and its image consists of the affine matrices whose translation coordinates are all even, so $[\bar{\Gamma} : \alpha(\bar{\Gamma})] = 8$. Translation a by the first standard basis vector lies outside $\alpha(\bar{\Gamma})$. So the preceding construction applies to $\bar{\Gamma}$, α and a.

Corollary 4.5 (the affine–Clifford trace)**.** *The affine–Clifford group W is sofic but not MF. Its canonical trace τ_W on $C^*_{\max}(W)$ is amenable but not quasidiagonal.*

Proof. The properties of $\bar{\Gamma}$ and α established above meet the hypotheses of Proposition 4.4, which gives every assertion. ☐

Proof of Theorem D. Proposition 4.4 identifies W with $K \rtimes \mathbb{Z}$ for a locally residually finite group K, and Corollary 4.5 gives the remaining assertions. ☐

5. A torsion-free finitely presented example

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. Hull's small cancellation theorem [Hu, Theorem 7.1] is stated with injectivity on a ball of $\Gamma(G, A)$, and a ball containing a given finite set gives the following form.

Theorem 5.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $g_1, \ldots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is acylindrically hyperbolic, $\varphi|_\Omega$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof treats $m = 1$ by passing to $G/\langle\langle W\rangle\rangle_G$ for one word W and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker\varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 5.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_\Omega$ is injective, and $\varphi(N) = Q$.*

Proof. The subgroup N is infinite and normal, so it acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1], and torsion-freeness makes it suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2\rangle$ is again suitable. Apply Theorem 5.1 to N_0 with $g_1, \ldots, g_m$ a finite generating set of G and with the set Ω. Then $Q = \langle\varphi(g_1), \ldots, \varphi(g_m)\rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$, so $Q = \varphi(N_0) = \langle\varphi(h_1), \varphi(h_2)\rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 5.1 and finitely presented because $\ker\varphi$ is normally generated by m elements. ☐

Fournier-Facio constructs a finitely presented torsion-free group G_0 with property (T), a subgroup $\Gamma \leq G_0$ with property (T), an element $t \in G_0$ with $t\Gamma t^{-1} \leq \Gamma$, and a subgroup $J \leq G_0$ isomorphic to a finitely presented infinite simple group, such that $[\Gamma, J] = 1$ and $tJt^{-1} \leq \Gamma$ [FF, §2]. The group G_0 is obtained there from Hull's common quotient theorem [Hu, Corollary 7.4], so it is acylindrically hyperbolic. Put $S = tJt^{-1}$. Since J is simple and nonabelian, J and S are perfect.

Lemma 5.3. *For every homomorphism $\rho\colon G_0 \to \bar{G}$,*

$$\rho(S) \leq \mathfrak{D}_{\bar{G}}(\rho(\Gamma)).$$

Proof. Write $\bar{\Gamma} = \rho(\Gamma)$, $\bar{t} = \rho(t)$, $\bar{J} = \rho(J)$, and $\bar{S} = \rho(S) = \bar{t}\bar{J}\bar{t}^{-1}$. Then $\bar{t}\bar{\Gamma}\bar{t}^{-1} \leq \bar{\Gamma}$, $\bar{J}$ centralizes $\bar{\Gamma}$, and $\bar{S} \leq \bar{\Gamma}$. For $c \in \bar{J}$ and $\ell \in \bar{S}$, the commutator $[\bar{t}c\bar{t}^{-1}, \ell]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$ by (1). As c ranges over $\bar{J}$, the element $\bar{t}c\bar{t}^{-1}$ ranges over $\bar{S}$, so $[\bar{S}, \bar{S}] \leq \mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. The group $\bar{S}$ is perfect as a quotient of S, so $\bar{S} \leq \mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. □

Proof of Theorem C. Let N be the normal closure of S in G_0, and fix $s \in S$ with $s \neq 1$. Lemma 5.2 applied to G_0, N, and the finite set $\{1, s\}$ gives $\varphi\colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi(N) = Q$, and $\varphi(s) \neq 1$. The group Q has property (T) as a quotient of G_0.

Lemma 5.3, applied directly to φ, gives $\varphi(S) \leq \mathfrak{D}_Q(\varphi(\Gamma))$. This defect subgroup is normal in Q, and the normal closure of $\varphi(S)$ is $\varphi(N) = Q$. Hence

$$\mathfrak{D}_Q(\varphi(\Gamma)) = Q.$$

Both Q and $\varphi(\Gamma)$ have property (T), as quotients of G_0 and Γ, respectively. The final clause of Theorem A therefore makes every homomorphism from Q to an MF group trivial. A homomorphism from a quotient of Q composes to one from Q, so no nontrivial quotient of Q is MF. □

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Use of AI and formal methods

Large language models were used extensively in mathematical exploration, proof development, manuscript drafting, literature navigation, Lean formalization, and editorial revision. The source markers link selected statements to Lean declarations; they do not certify every proof in this revision. No fresh comprehensive Lean build or dependency audit is claimed here, and the cited literature remains an external input.

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